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RESEARCH ARTICLE

The effects of river barriers and range expansion of the population genetic structure and stability in Greenside Darter (*Etheostoma blennioides*) populations

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Abstract The genetic structure of a stream-dwelling fish, the Greenside Darter, *Etheostoma blennioides,* is described from variation at nine microsatellite loci in 26 populations in the northern-most portions of the species' range in southern Ontario, Canada in two sampling years. We found relatively high levels of genetic structure at the among- and within-watershed scales, with some watersheds and populations exhibiting very high divergence. The Ausable River populations were especially isolated, containing distinct populations of potential conservation concern. Temporal replicates at selected localities showed evidence of substantial temporal variation in genetic structure, perhaps resulting from movement among sites. We found strong evidence for an effect of river barriers (dams and weirs) on dispersal measured by genotype assignment techniques. However, we found no bias in upstream vs downstream dispersal. Significant isolation-by-distance relationships in both sample years indicate that river distance is an important factor regulating gene flow in these watersheds. The Canadian Greenside Darter populations are expanding their range into more northerly watersheds, but also show substantial within-watershed genetic structure despite substantial dispersal.

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Introduction

There are two principal explanations for the location of species distribution boundaries. One is that the species is in the process of dispersing and colonizing (range expansion) and, thus, the current distribution boundary is a temporary artifact of a non-equilibrium process (e.g., a newly introduced invasive species exploiting a novel habitat; Ricciardi 2003). The second possibility is that the species distribution is at equilibrium, and the organisms at the range-edge are at their acclimation/adaptation limit and cannot successfully colonize beyond that boundary (Fahrig and Merriam 1994). In the northern hemisphere, range boundaries are typically extending north, while extinctions are occurring along the southern boundaries (Arnell and Reynard 1996; Magnuson et al. 1997). Therefore, populations at their northern boundaries may be those most suited to establishing themselves in the new habitats created by warming temperatures (Fraser 2000). In general, populations at the edges of geographic ranges may play a role in survival through long-term environmental change (Hunter 1991; Quinn and Karr 1993). However, as the distance from the center of the species distribution increases, gene flow typically decreases, facilitating genetic divergence and loss of genetic diversity by genetic drift (Bunnell et al. 2004). Such a pattern has been documented in a variety of species (Lammi et al. 1999; Kyle and Strobeck 2002; Hutchinson 2003; Bouzat and Johnson 2004).

The Greenside Darter, *Etheostoma blennioides*, is a widely distributed, stream-dwelling fish in the eastern

United States, with its most northern boundary in southern Canada. About 10,000 years ago, at the time of the Wisconsin deglaciation, this species migrated north from the Mississippian refugium into the glacial Lake Maumee (ancestor of the present-day Lake Erie) (Mandrak and Crossman 1992). This Mississippian population expanded its range west by spreading through the Michigan basin and managed to colonize only a few watersheds in southern Ontario (Fig. 1). In the 1980s the Greenside Darter had decreased in number to the extent that it was listed as a 'Special Concern' species by the Committee On the Status of Endangered Wildlife In Canada (COSEWIC; Dalton 1991). In 2006, the species was re-evaluated and determined to be increasing in numbers and expanding its range such that it was no longer considered a species at risk (COSEWIC http://www.cosewic.gc.ca, 2006). Also, in the early 1990s, the species was first collected in the Grand River watershed in Ontario, adding to the three Canadian watersheds (Thames, Sydenham, and Ausable Rivers) known to have Greenside Darter. The Greenside Darter in Ontario provides a good opportunity for evaluating genetic relationships among populations of conservation interest due to past declines and recent colonizations.

There are currently no known distinctions among the northern populations of the Greenside Darter that warrant conservation designation below the species level (Dalton 1991); however, there have been no genetic studies done on these populations. Previous studies based on allozyme electrophoresis and mtDNA sequences on populations of Greenside Darter in the United States showed that gene flow varies among regions studied (Heithaus and Laushman 1997; Turner and Trexler 1998). Faber and White (2000) found no significant divergence between populations in two unconnected Ohio River tributaries, whereas Turner (2001) found populations within a 66-km study area in the Spring River, Arkansas were significantly divergent.

We used microsatellite DNA variation to assess population structure and dynamics among all Greenside Darter populations in Canada. We tested for temporal stability and measured population genetic structure and dispersal. We also tested for within-watershed barrier effects on gene flow patterns, as well as within-watershed dispersal. Our analyses characterize the genetic relationships among the isolated and fragmented populations of a riverine fish at their northern species range boundary. This study identifies isolated river and watershed populations as potential units for conservation and, more generally, it provides new insight into connectivity and barriers to gene flow among populations at the within- and among-watershed scales.

Methods

Survey sampling

We sampled Greenside Darters in Canada in 2005 and 2006 to determine the current distribution of Greenside Darters in Southern Ontario, and to collect fin clips for genetic analyses. To determine the current distribution of the Greenside Darter we targeted rivers known to support the species, and we sampled rivers and lake shores immediately surrounding the known distribution. Sites sampled were selected based on the preferred habitat of the Greenside Darter: moderate flow, gravel to boulder substrate, low turbidity and presence of riffles. Most sites were systematically single-pass electroshocked using a Smith-Root LR-24 backpack electrofisher, with an average effort of 1800 shocking seconds per site. Where electrofishing would not be effective (shores of lakes

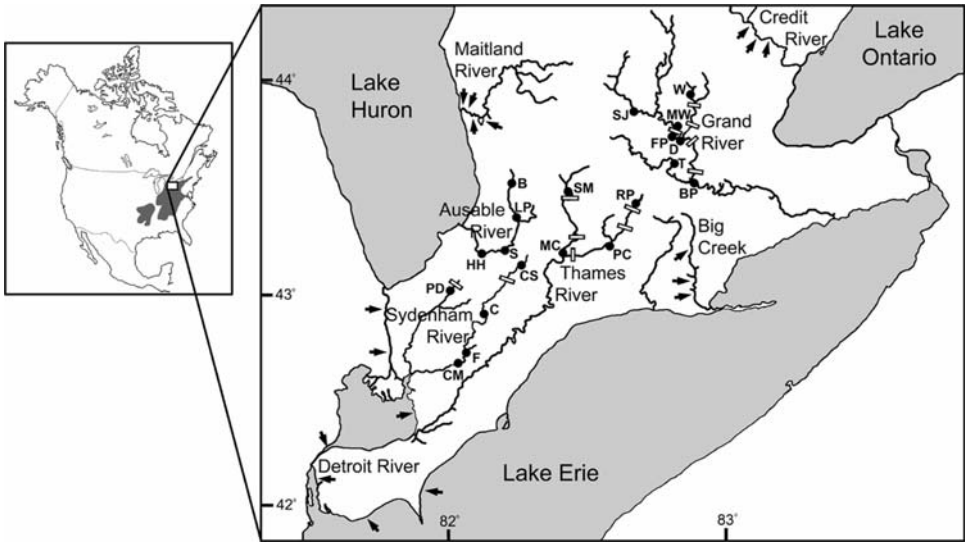

Fig. 1 Map showing the sample sites for the Greenside Darters (*Etheostoma blennioides*) within four watersheds in southern Ontario, Canada. The species distribution of Greenside Darters (dark grey shading) is shown on the inset map of North America. Sampled sites are identified by filled circles, while sampling sites that yielded no Greenside Darters are indicated by arrows. River barriers are indicated by open bars across rivers

and large rivers), sites were seined 3–10 times with a 6-m beach seine with 2.5-mm mesh. The species was considered absent at a particular site when it was not collected within the first 1000 shocking seconds or within three passes of the beach seine.

Sample collection

In October 2005, Greenside Darters were collected from 14 sites in the Ausable, Grand and Sydenham rivers in southwestern Ontario. In June 2006, they were collected at six new sites: one each in the Ausable and Sydenham rivers, and four in the Thames River (Fig. 1). To evaluate the temporal stability of the genetic structure of the Greenside Darter populations, six sites, three each in the Ausable and Sydenham rivers, were sampled in 2005 and re-sampled in 2006. A small caudal fin clip was taken from each captured fish and stored in 95% ethanol for later DNA extraction.

DNA extraction and genotyping

DNA was extracted from 1352 Greenside Darter clips by using either Promega® Wizard kit salt-based extraction, or column-based plate extraction (Elphinstone et al. 2003) methods. To ensure allele sizes and resolution were unaffected by extraction method, 64 fin clips were extracted using both methods, and no genotype anomalies resulted. All individuals were genotyped at six *E. blennioides* microsatellite loci (one tetra-nucleotide and five di-nucleotide repeat markers; Ebl4, Ebl1, Ebl3, Ebl5, Ebl7 and Ebl8) from Beneteau et al. (2007), and at three tetra-nucleotide microsatellite loci designed for *Etheostoma caeruleum* (Eca10, Eca11 and Eca48) from Tonnis (2006). Each locus was amplified in 10-µl polymerase chain reactions (PCR) containing approximately 50-ng template DNA, 32-µM dye-labelled forward primer, 0.5-µM reverse primer, 200-µM of each dNTP, 2.5-mM $MgCl_2$, and 0.5-U *Taq* DNA polymerase (Applied Biosystems, Foster City, USA) in a 1X PCR buffer. Amplification employed the following thermal protocol: 2 min at 95°C; 30 cycles of 15 s at 95°C, 15 s at 56°C, 30 s at 72°C, followed by a 2 min extension at 72°C and a 4°C hold. Allele sizes were scored using a LiCor 4300 DNA Analyzer and the software GENE IMAGIR 4.05 (Scanalytics, Inc.).

Microsatellite loci were tested for linkage disequilibrium and Hardy-Weinberg equilibrium using ARLEQUIN v3.0 (Schneider et al. 2000), with significance adjusted for multiple tests using sequential Bonferroni correction. We also tested for null alleles, large allele drop-out, etc. using MICROCHECKER v2.2.3 (Van Oosterhout et al. 2004). Allelic richness and number of private alleles at each sampling site were calculated using FSTAT v.2.9.3 (Goudet 2001).

Temporal stability

We first tested for temporal stability to determine if our data from 2005 and 2006 could be combined for the locations sampled in both years; these included three sites each from the Ausable (LP, B and HH; Fig. 1) and Sydenham (F, C and PD; Fig. 1) watersheds. Exact tests were used to compare the allele frequencies of each sampling site in 2005 and 2006 (10,000 dememorizations with 20,000 permutations; TFPGA v1.3; Miller 1997). The temporal variation of genetic diversity was examined by comparing expected heterozygosity (H_E) and allelic richness (A) of the temporal samples of each site. We calculated F_{ST} (Weir and Cockerham 1984) values between years within each site, and we performed a nested hierarchical analysis of molecular variance (AMOVA; 10,000 permutations) to partition genetic diversity attributable to differences among years (2005 and 2006) from that attributable to differences among sampled sites using ARLEQUIN v3.0.

Genetic structure analyses

Among watersheds

To evaluate among-watershed genetic divergence, we calculated pairwise F_{ST} among watersheds by combining genotype data across all sample sites within each watershed. Pairwise chord distances (D_C; Cavalli-Sforza and Edwards 1967) were computed among all samples and a neighbor-joining cluster analyses with 10,000 bootstraps over loci in POPULATIONS v1.2.28 (Langella 2002) was performed. Data from both sampling years and all sample sites were included to allow comparisons among years as well as among sites. Chord distance provides accurate tree topology for closely related populations (Angers and Bernatchez 1998). Trees were prepared with TREEVIEW (Win32; Page 2001). AMOVAS (10,000 permutations) were used to quantify molecular genetic variance explained by among- versus within-watershed variation in the two sampling years.

Within watersheds

We computed pairwise F_{ST} and D_C values between sampling sites (for both years separately) within watersheds, and used TFPGA v1.3 to perform pairwise exact tests of allele frequency differences (Raymond and Rousset 1995) among populations.

Isolation-by-distance

We performed Mantel tests for correlation between the matrix of shortest pairwise water distances among sample sites (measured using Google Earth) and the matrix of

pairwise D_C values using GENALEX 6.0 with 9999 permutations (Peakall and Smouse 2006). Due to significant temporal variation in genetic structure (see Results), we performed these isolation-by-distance (IBD) analyses for 2005 and 2006 samples separately.

Dispersal (within watersheds)

To determine dispersal patterns, we performed genotype assignment across all genotypes within each watershed, within sample year, using the partial Bayesian method of Rannala and Mountain (1997) with a 95% assignment threshold with GENECLASS 2.0 (Piry et al. 2004). For fish having at least one site assignment above the 95% threshold, we identified the most likely source site using the rank-based assignment method with the criterion that the ratio of the highest ranked likelihood score of assignment to the second-most likely score must exceed 4 for the individual to be successfully assigned. Dispersing fish (migrants) were thus those assigned to sites other than the site of capture. All identified migrant paths were identified as "upstream versus downstream" and "across a barrier versus no barrier" dispersal events, where barriers were identified by local conservation authorities as dams or weirs that altered water low in the river or stream. We tested for direction and barrier-crossing biases in fish dispersal by χ^2 analyses using random dispersal as the null expectation. Thus for the upstream-downstream comparison, the null expectation was a 1:1 ratio, while the null expectation for the barrier-no barrier comparison depended on the ratio of possible dispersal pathways with barriers versus those with no barrier (see Fig. 1 for barrier distribution relative to sampling sites). The dispersal data from both years were combined for the χ^2 analyses, although the assignment analyses were restricted to genotype data within a single year.

Results

Survey sampling

Sampling in 2005 and 2006 to determine the Canadian distribution of *Etheostoma blennioides* in the streams, rivers and lake shores immediately surrounding the known distribution resulted in no additional locations (see Fig. 1) over those reported in the most recent published distribution map (COSEWIC http://www.cosewic.gc.ca, 2007).

Genotyping

All nine microsatellite loci were highly variable (7 to 70 alleles per locus). One locus (Ebl1) was fixed in all Ausable sites sampled in 2005 and 2006 except for Lyon's Park in 2006 (Table 1). Departures from Hardy-Weinberg equilibrium were observed at 3 out of 234 locus-by-population combinations [(HH ('05) at *Ebl*; at *Ebl5*; LP] ('06) at *Ebl7*). There was no evidence of linkage disequilibrium and there was no evidence of null alleles, band calling errors, or large-allele drop out based on MICROCHECKER v2.2.3 analyses.

Temporal stability

Five of six sites sampled in 2005 and re-sampled in 2006 showed significant differences in allele frequency distributions between years, the exception was PD in the North Sydenham River (Fig. 2). There was no temporal difference in H_E and A in the populations sampled in 2005 and 2006. Generally, F_{ST} values across the two sampling years grouped into two categories; the more geographically isolated sites (North Sydenham River site, PD, and Ausable watershed sites) were not significantly different from zero, whereas the two sites in the main East Sydenham River (F and C) had statistically significant F_{ST} values (Table 2; Fig. 2). The AMOVA attributed 93.1% ($P < 0.0001$) of the genetic diversity to within-site variation, 4.8% ($P < 0.01$) to variation among-sites, and 2.0% ($P < 0.0001$) to variation between-years, within sites.

Genetic structure

Among watersheds

The mean pairwise F_{ST} among watersheds was 0.079 (2005) and 0.055 (2006; Table 3). The neighbor-joining tree resolves four distinct groups with moderate bootstrap support (Fig. 3). These groups clearly correspond to the watersheds, with the Thames and Sydenham watersheds grouping separately from the other watersheds. Interestingly, the temporal replicated samples did not tend to cluster together, although they all clustered within the correct watershed (Fig. 3). The long within-cluster branch lengths suggest further geographic-related structure within watersheds. The AMOVA using the 2005 data attributed 91.5% ($P < 0.0001$) of the genetic diversity to within-site variation among individuals, 7.2% ($P < 0.0001$) to variation among-watersheds, and 1.3% ($P < 0.0001$) to variation among-sites, within watersheds. The AMOVA using the 2006 data attributed 94.3% ($P < 0.0001$) of the genetic diversity to within-site variation among individuals, 5.2% ($P < 0.0001$) to variation among-watersheds, and 0.5% ($P < 0.001$) to variation among-sites, within watersheds.

Within watersheds

Exact tests of differences in allele frequency distribution were significant for most pairwise comparisons of sites

Table 1 Site name, letter code (see Fig. 1) and coordinates, as well as the number of *E. blennioides* caught at each site by sampling year

Watershed	Site name	Code	Lat., Long.	Year	*N*	Mean H_O	Mean H_E	Mean A	Total A_P
Sydenham	Florence	F	42°38′ N, 82°00′ W	2005	59	0.65	0.69	13	5
	Cider Mill	CM	42°36′ N, 82°04′ W	2005	68	0.66	0.69	13	7
	Carolinian	C	42°46′ N, 81°50′ W	2005	64	0.67	0.74	13	3
	Petrolia Dam	PD	42°52′ N, 82°08′ W	2005	27	0.60	0.67	9	2
	Florence	F	42°38′ N, 82°00′ W	2006	50	**0.67**	0.69	12	5
	Carolinian	C	42°36′ N, 82°04′ W	2006	49	0.68	0.68	13	5
	Petrolia Dam	PD	42°52′ N, 82°08′ W	2006	19	0.65	0.71	9	1
	Coldstream	CS	43°00′ N, 81°30′ W	2006	31	0.70	0.72	10	0
Ausable	Lyon's Park	LP	43°08′ N, 81°32′ W	2005	46	0.58	0.63	6	0
	Brinsley	B	43°12′ N, 81°31′ W	2005	90	0.63	0.63	7	0
	Hungry Hollow	HH	43°04′ N, 81°47′ W	2005	52	**0.62**	0.66	8	5
	Lyon's Park	LP	43°08′ N, 81°32′ W	2006	46	**0.56**	0.62	6	1
	Brinsley	B	43°12′ N, 81°31′ W	2006	28	0.57	0.62	6	0
	Hungry Hollow	HH	43°04′ N, 81°47′ W	2006	64	0.56	0.63	7	2
	Springbank	S	43°03′ N, 81°41′ W	2006	27	0.58	0.63	6	0
Grand	Bean Park	BP	43°10′ N, 80°22′ W	2005	74	0.65	0.69	8	0
	St. Jacob's	SJ	43°32′ N, 80°33′ W	2005	102	0.68	0.71	9	0
	Trussler	T	43°17′ N, 80°28′ W	2005	116	0.70	0.71	9	1
	Woodlawn	W	43°34′ N, 80°16′ W	2005	65	0.66	0.68	7	0
	Doon	D	43°23′ N, 80°23′ W	2005	42	0.69	0.69	7	0
	Freeport	FP	43°24′ N, 80°26′ W	2005	60	0.67	0.71	8	1
	Mannheim Weir	MW	43°25′ N, 80°25′ W	2005	40	0.69	0.71	8	0
Thames	St. Mary's	SM	43°17′ N, 81°09′ W	2006	22	0.71	0.73	11	1
	Roth Park	RP	43°08′ N, 80°45′ W	2006	52	0.72	0.73	14	4
	Piney Creek	PC	42°58′ N, 80°57′ W	2006	28	0.71	0.74	12	2
	Medway Creek	MC	43°00′ N, 81°16′ W	2006	31	0.74	0.72	11	1

Observed heterozygosity (H_O), expected heterozygosity (H_E), and number of alleles (A) averaged over nine microsatellite loci, and total number of private alleles (A_P) calculated for all sampled sites in 2005 and 2006

Populations out of Hardy-Weinberg equilibrium are indicated with the H_O in bold type

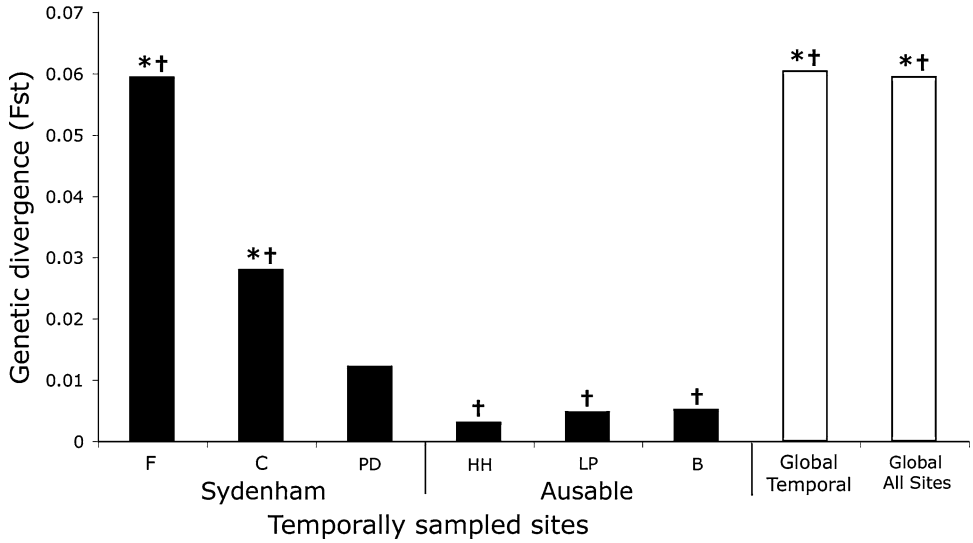

Fig. 2 F_{ST} values for Greenside Darters (*E. blennioides)* between temporal replicates at six sites (labeled as in Fig. 1) sampled in 2005 and 2006. Global F_{ST} for all temporally sampled sites and for all data combined are presented as white bars for comparison. Significant F_{ST} values are indicated with an asterisk and significant differences in allele frequency distribution are indicated with a dagger

Table 2 Within-watershed sampling site comparison of genetic distance (D_C; above diagonal) and Weir and Cockerham's F_{ST} (below diagonal)

Sydenham	F 05	F 06	C 05	C 06	PD 05	PD 06	CM 05	CS 06
F 05	–	**0.27**	**0.23**	**0.27**	**0.37**	**0.38**	**0.20**	**0.28**
F 06	**0.055**	–	**0.25**	0.21	**0.31**	**0.31**	**0.25**	**0.24**
C 05	**0.012**	**0.022**	–	**0.25**	**0.34**	**0.35**	**0.23**	**0.26**
C 06	**0.053**	0.003	**0.024**	–	**0.29**	**0.31**	**0.26**	**0.25**
PD 05	**0.043**	**0.007**	**0.012**	**0.012**	–	0.29	**0.37**	**0.34**
PD 06	**0.067**	**0.015**	**0.030**	**0.008**	**0.023**	–	**0.38**	**0.34**
CM 05	0.005	**0.053**	**0.017**	**0.058**	**0.051**	**0.074**	–	**0.29**
CS 06	**0.071**	**0.016**	**0.037**	**0.013**	**0.027**	0.005	**0.079**	–

Ausable	LP 05	LP 06	B 05	B 06	HH 05	HH 06	S 06
LP 05	–	**0.18**	**0.12**	**0.18**	**0.20**	**0.15**	**0.20**
LP 06	0.003	–	**0.15**	**0.16**	**0.23**	**0.15**	0.17
B 05	**0.007**	**0.006**	–	**0.16**	**0.18**	**0.13**	0.16
B 06	**0.010**	0.006	0.005	–	**0.24**	0.15	0.20
HH 05	**0.013**	**0.013**	0.004	**0.012**	–	**0.19**	**0.19**
HH 06	**0.008**	**0.008**	**0.006**	0.006	0.005	–	0.15
S 06	0.004	0.002	−0.005	−0.002	−0.001	−0.006	–

Grand 2005	BP	SJ	T	W	D	FP	MW
BP	–	**0.15**	**0.13**	**0.16**	0.15	**0.17**	**0.20**
SJ	0.004	–	**0.16**	**0.21**	**0.16**	**0.18**	0.16
T	0.002	**0.009**	–	**0.17**	**0.16**	**0.16**	**0.20**
W	**0.012**	**0.024**	**0.014**	–	**0.18**	**0.20**	**0.24**
D	0.000	0.004	0.005	**0.007**	–	0.16	**0.21**
FP	0.004	0.003	**0.007**	**0.016**	0.004	–	**0.19**
MW	0.008	0.003	**0.011**	**0.022**	0.013	0.001	–

Thames 2006	SM	RP	PC	MC
SM	–	**0.29**	**0.31**	0.26
RP	**0.008**	–	**0.28**	**0.25**
PC	0.005	0.003	–	**0.27**
MC	0.005	**0.007**	−0.005	–

Bold type indicates statistical significance: above the diagonal, results from pairwise exact tests for allele frequency differences are shown, below the diagonal, significance of the F_{ST} values are shown. Site abbreviations are as in Table 1 and Fig. 1, and temporal sites are identified by year (05 or 06)

Table 3 Mean pairwise F_{ST} values among populations of Greenside Darters from four watersheds sampled over two year (2005 and 2006)

	Sydenham	Ausable
Ausable	0.099 ('05); 0.094 ('06)	–
Grand	0.058 ('05)	0.091 ('05)
Thames	0.016 ('06)	0.064 ('06)

The Thames and Grand Rivers were not sampled in the same year; however the Sydenham and Ausable Rivers were sampled in both years (see Table 1)

within watersheds, with close proximity between sites appearing to be the main exception (Table 2). However, in the Ausable watershed in 2006, only one of the three pairwise site comparisons showed significant allele frequency differences. Patterns of genetic divergence (F_{ST}) among sites varied by watershed and year, but generally were significant and reflected the degree of spatial separation (Table 2).

Isolation by distance

The Mantel test of IBD among the 2005 samples was significant ($P = 0.010$; $R^2 = 0.62$); however inspection of the scatterplot identifies an anomalous grouping of points that corresponds to the among-watershed comparisons involving the Grand River (Fig. 4a). The Grand River Greenside Darter are believed to be recently introduced, and thus their relatively low genetic divergence relative to their geographic distance from the other sampled sites (Fig. 4a) may reflect anthropogenic effects. When we exclude the Grand River data, and perform the Mantel test

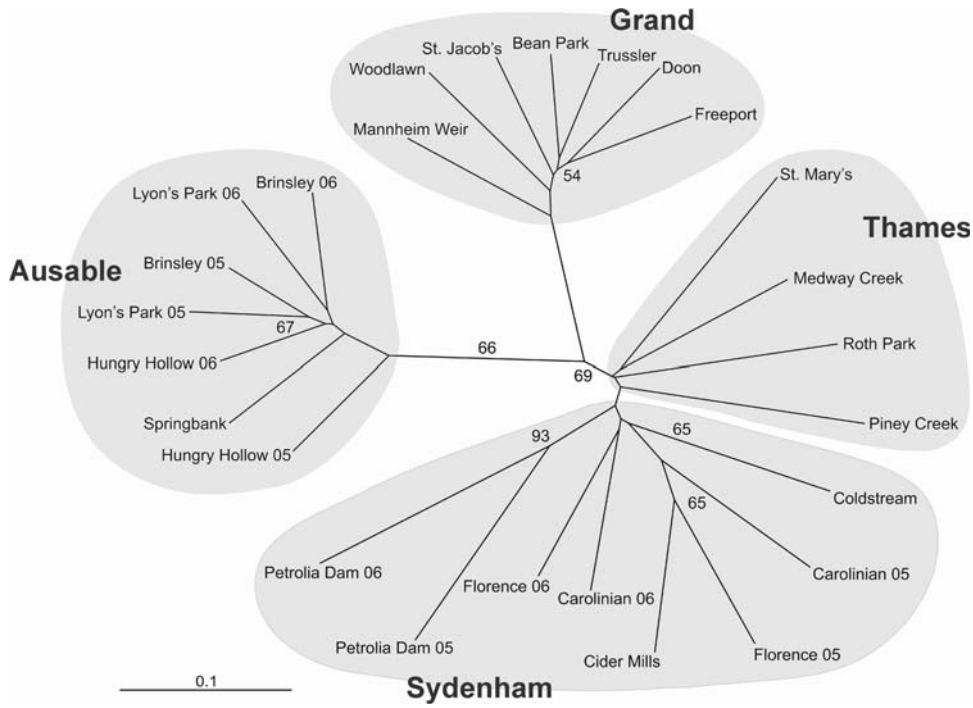

Fig. 3 Neighbor-joining cluster diagram (D_C) of all samples of Canadian Greenside Darter populations. Numbers indicate replicated bootstrap values out of 10,000. Watershed associations indicated by grey shading

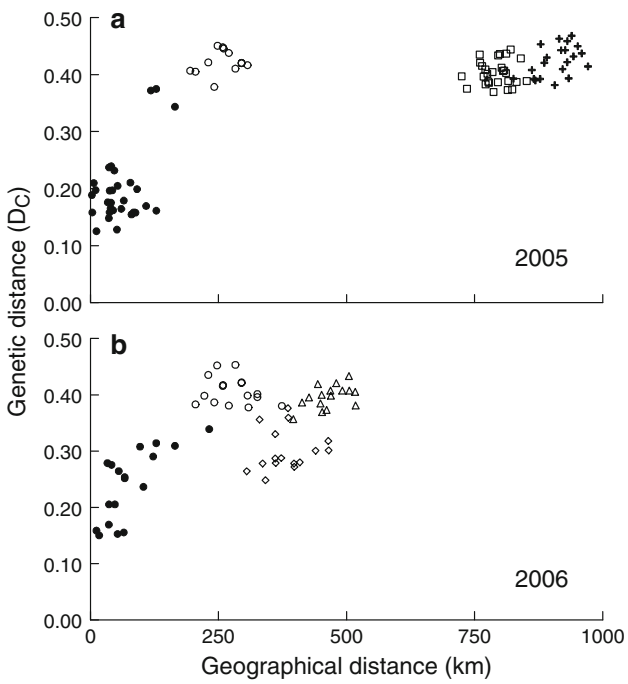

Fig. 4 Scatterplot of pairwise *E. blennioides* genetic distance (D_C) against shortest water distance (km) among sites for the two sampling years separately. *Panel a*: Sites sampled in 2005; filled circles (●) are within-watershed comparisons, all others are among-watershed comparisons (○ = Sydenham-Ausable; □ = Sydenham-Grand; + = Ausable-Grand). *Panel b*: Sites sampled in 2006; filled circles (●) are within watershed comparisons, all others are among-watershed comparisons (○ = Sydenham-Ausable; ◇ = Sydenham-Thames; Δ = Ausable-Thames). Significant isolation-by-distance relationships are present in both years, as well as in 2005 with Grand River data removed (see text)

for IBD among the Sydenham and Ausable River samples, a significant positive relationship emerges ($P < 0.01$; $R^2 = 0.84$). The IBD analysis using the 2006 samples yielded a significant positive relationship based on a Mantel test ($P = 0.02$; $R^2 = 0.41$). IBD analysis within each watershed by sample year showed no significant correlation between geographic and genetic distance by Mantel tests, likely due to low sample sizes (e.g., the Ausable 2005 had only three comparisons).

Dispersal (within watershed)

In all watersheds, dispersal (determined by genotype assignment) among sites was highest at shorter distances, with the exception of the Grand, which showed the highest number of migrants between sites 121–140 km apart (Fig. 5). The total number of individuals moving among sites at any distance was considerably lower in the Ausable watershed than all other watersheds in either year. We found no significant difference in the frequency of upstream versus downstream dispersal (Table 4) in any watershed. Only two watershed-year combinations had both barrier and no-barrier dispersal categories for comparison (Sydenham 2006; Grand 2005; Table 4). In both cases we found that the frequency of across-barrier dispersal was significantly lower than random expectation, while the frequency of no-barrier dispersal events was higher than random expectation (Sydenham: $\chi^2 = 19.0$, df = 1; Grand: $\chi^2 = 22.5$, df = 1; Table 4).

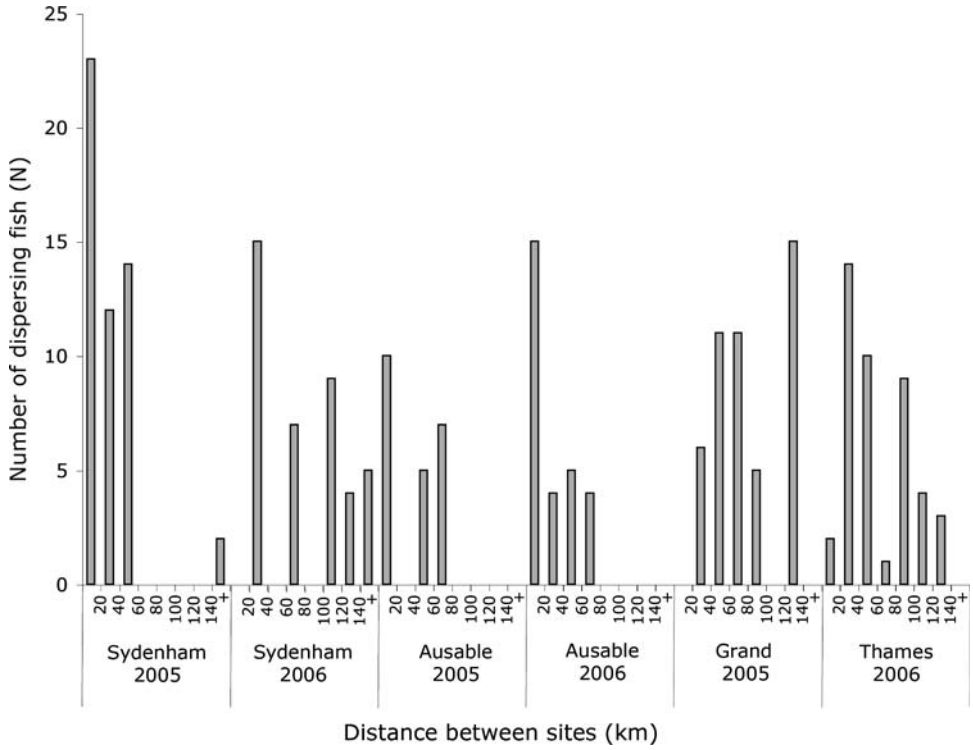

Fig. 5 Frequency distribution of within-watershed dispersal distances of *E. blennioides*. Distance is the shortest water distance between the capture site and the genotype assigned site of origin

Table 4 The number of *E. blennioides* migrants within each watershed, separated by sampling year, dispersing either upstream/downstream, or across a barrier (dam or weir) or not (a dash indicates that no pathway among sample sites fulfilled the criterion)

Dispersal	Sydenham		Ausable		Grand	Thames
	2005	2006	2005	2006	2005	2006
Downstream	28 (26.5)	23 (23.5)	13 (11)	17 (12.5)	20 (26.5)	37 (32.5)
Upstream	25 (26.5)	24 (23.5)	9 (11)	8 (12.5)	33 (26.5)	28 (32.5)
No barrier	53	30 (15.6)	22	25	18 (6.6)	–
Across barrier	–	17 (31.4)	–	–	35 (46.4)	65

Numbers in parentheses are the expected numbers of dispersers based on random dispersal among the sampled sites

No significant differences were found for upstream vs downstream; however there is significant and strong bias against across-barrier dispersal

Discussion

Populations at the boundary of their species range typically exhibit genetic structure that reflects their relatively isolated and fragmented distribution (Bunnell et al. 2004; Lesica and Allendorf 1995). Greenside Darter populations in Canada are at the northern-most edge of their geographic range, and our results are consistent with expectations for peripheral populations, although our data do not provide a direct comparison with central-range populations. Generally, the Canadian watersheds containing Greenside Darter were isolated, based on watershed-level F_{ST} values and AMOVA results; however some watershed comparisons appear to reflect recent and possibly on-going gene flow (e.g., Sydenham and Thames). Such levels of differentiation likely reflect long-term isolation and, given the level of genetic diversity within and among watersheds, the northern-range Greenside Darter harbour important genetic resources that may contribute to the adaptive potential of the species.

The Ausable watershed populations, which are the most northern Greenside Darter, show very little spatial genetic structure, short-distance within-watershed dispersal and low levels of genetic diversity. This might reflect either (1) recent re-colonization and spread in the Ausable watershed after the early 1990 s when Greenside Darter were thought to be extirpated from the system (Dalton 1991), or (2) population expansion subsequent to very low population sizes in the early 1990s. In either case, the Ausable populations are expanding. This rapid recovery may reflect a pattern of northward range expansion. For example, extensive sampling of the Maitland River watershed in 2005

and 2006 produced several other darter species, but no Greenside Darter, whereas, in the fall of 2007, electrofishing at the same sites in the Maitland River for another project yielded 30–50 Greenside Darter (C. Beneteau, unpublished data). This is the first record of this species in the Maitland River watershed.

Greenside Darter populations in the Thames and Sydenham watersheds were weakly divergent, possibly as a result of on-going gene flow through Lake St. Clair where the species has been observed along the shore between the two rivers (Dalton 1991; COSEWIC http://www.cosewic.gc.ca, 2007). Another possibility is gene flow via headwater connections, since the upper watersheds are separated by short land distance but considerable water distance. The latter possibility is supported by the pattern of among-watershed population divergences: The sample site closest to the mouth of the Sydenham River (CM) has an average pairwise F_{ST} divergence from the Thames sites of 0.059 (±0.004 SEM), while the sample site furthest from the mouth of the Sydenham River (CS) has an average pairwise F_{ST} value of 0.0088 (±0.003 SEM). The relatively low divergence between populations in the Sydenham and Thames River watersheds is thus likely the result of historic connection and headwater migration or, alternatively, human mediated movement.

The temporally sampled Greenside Darter populations generally showed remarkable levels of genetic instability with five of six populations exhibiting significant changes in allele frequencies between 2005 and 2006, and with a significant proportion of the observed genetic variation attributed to temporal variation. Populations on the periphery of a species distribution are expected to be temporally and genetically unstable, primarily due to their use of sub-optimal habitats and resulting small population sizes and genetic drift (Hoffman and Blows 1994) and limited connectivity to other populations (Kyle and Strobeck 2002). However, the levels of change documented here are difficult to attribute to drift, and more likely reflect fish movements. Possibilities include seasonal migration or dispersal effects, since sites were sampled in October in 2005, and in June in 2006. The sampled sites exhibited reduced genetic divergence between neighboring sites in June (2006) relative to October (2005). Spawning in northern populations of Greenside Darter begins in April and continues into June (Scott and Crossman 1973), and our June results might be affected by spawning-related movement. Greenside Darters may be less sedentary than expected, and may disperse long distances or occupy larger home ranges in the early summer or during the spawning months. Reasons why darters might show seasonal movement patterns include minimization of metabolic requirements, escape of environmental extremes, and maximization of foraging or breeding opportunities (Schlosser and Toth 1984; Cunjack 1988; Labbe and Fausch 2000). Further sampling would be required to determine if the temporal instability documented here is the result genetic instability in the Canadian populations, or an artifact of seasonal dispersal/migration.

Contrary to previous studies on Greenside Darter dispersal (Winn 1958; Bunt et al. 1998), we found no evidence of direction-biased (upstream vs downstream) movement within any of the sampled watersheds in either year. However, dispersal was affected by the presence of dams or weirs. Although the magnitude of the fish movement disruption likely varies among the identified barriers, it is clear that as a group they represent substantial dispersal barriers, at least within the Sydenham and Grand Rivers. In the Grand River watershed a substantial component of all dispersal events were upstream and over a barrier (although fewer than expected across barriers); however, the Grand River is a newly colonized river and patterns of dispersal identified here may reflect introduction effects rather than natural dispersal. In general, dispersal within watersheds is primarily driven by unknown factors, possibly seasonally variable factors, although there is a short-dispersal distance bias in most watersheds. The effect of river distance on gene flow is reflected in the isolation-by-distance relationships, where geographic distance explains approximately 40–85% of the observed variation in genetic divergence. However, our IBD analysis included among-watershed comparisons, thus the relationships identified may include colonization effects, rather than simple migration-drift equilibrium. A better understanding of dispersal in Greenside Darter is likely to be of considerable value in predicting future population dynamics and changes in species range.

We found no genetic evidence for Canadian populations that warrant special conservation designation (Dalton 1991), but there is substantial genetic structure, especially among watersheds. The Ausable watershed populations appear to have experienced substantial bottlenecks and exhibit reduced genetic diversity relative to other Canadian populations, although those populations are rapidly recovering. The pattern of recovery, novel habitat colonization, and rapid population growth of the Canadian Greenside Darter may reflect changes in the northern habitats that are allowing general range expansion. Our work characterizes the complex and temporally unstable genetic relationships among the Canadian populations of Greenside Darter, and indicates that their conservation, although simplified by their increasing numbers and expanding range, may not be straightforward.

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References

Angers B, Bernatchez L (1998) Combined use of SMM and non-SMM methods to infer fine structure and evolutionary history of closely related Brook Charr (*Salvelinus fontinalis*, Salmonidae) populations from microsatellites. Mol Biol Evol 15:143–159

Arnell NW, Reynard NS (1996) The effects of climate change due to global warming on river flows in Great Britain. Hydrology 183:397–424. doi:10.1016/0022-1694(95)02950-8

Beneteau CL, Mandrak NE, Heath DD (2007) Characterization of eight polymorphic microsatellite DNA markers for the Greenside Darter, *Etheostoma blennioides* (Percidae). Mol Ecol Notes 7:641–643. doi:10.1111/j.1471-8286.2006.01661.x

Bouzat JL, Johnson K (2004) Genetic structure among closely spaced leks in a peripheral population of lesser prairie chickens. Mol Ecol 13:499–505. doi:10.1046/j.1365-294X.2003.02068.x

Bunnell FL, Campbell RW, Squires KA (2004) Conservation priorities for peripheral species: the example of British Columbia. Can J Res 34:2240–2247. doi:10.1139/x04-102

Bunt CM, Cooke SJ, McKinley RS (1998) Creation and maintenance of habitat downstream from a weir for the Greenside Darter, *Etheostoma blennioides* – a rare fish in Canada. Environ Biol Fishes 51:297–308. doi:10.1023/A:1007444811400

Cavalli-Sforza LL, Edwards AWF (1967) Phylogenetic analysis: models and estimation procedures. Evol Int J Org Evol 21: 550–570. doi:10.2307/2406616

Cunjack RA (1988) Physiological consequences of over-wintering in streams: the cost of acclimatization. Can J Fish Aquat Sci 45:443–452

Dalton KW (1991) Status of the Greenside Darter, *Etheostoma blennioides*, in Canada. Can Field Nat 105:173–178

Elphinstone MS, Hinten GN, Anderson MJ, Nock CJ (2003) An inexpensive and high-throughput procedure to extract and purify total genomic DNA for population studies. Mol Ecol Notes 3:317–320. doi:10.1046/j.1471-8286.2003.00397.x

Faber JE, White MM (2000) Comparison of gene flow estimates between species of darters in different streams. J Fish Biol 57:1465–1473. doi:10.1111/j.1095-8649.2000.tb02225.x

Fahrig L, Merriam G (1994) Conservation of fragmented populations. Conserv Biol 8:50–59. doi:10.1046/j.1523-1739.1994.08010050.x

Fraser DF (2000) Species at the edge: the case for listing of "peripheral" species. In: Darling LM (ed) Proceedings of a conference on the biology and management of species and habitats at risk, Kamloops, 1999

Goudet J (2001) FSTAT, a program to estimate and test gene diversities and fixation indices (version 2.9.3). Department of Ecology & Evolution, University of Lausanne, Switzerland. See http://www2.unil.ch/popgen/softwares/fstat.htm. Accessed 30 Nov 2007

Heithaus MR, Laushman RH (1997) Genetic variation and conservation of stream fishes: influence of ecology, life history, and water quality. Can J Fish Aquat Sci 54:1822–1836. doi:10.1139/cjfas-54-8-1822

Hoffman AA, Blows MW (1994) Species borders: ecological and evolutionary perspectives. Trends Ecol Evol 9:223–227. doi:10.1016/0169-5347(94)90248-8

Hunter ML Jr (1991) Coping with ignorance: the coarse-filter strategy for maintaining biodiversity. In: Holm K (ed) Balancing on the brink of extinction: the Endangered Species Act and lessons for the future. Island Press, Washington, DC

Hutchinson DW (2003) Testing the central/peripheral model: analyses of microsatellite variability in the eastern collared lizard (*Crotaphytus collaris collaris*). Am Midl Nat 149:148–162. doi:10.1674/0003-0031(2003)149[0148:TTCPMA]2.0.CO;2

Kyle CJ, Strobeck C (2002) Connectivity of peripheral and core populations of North American wolverines. J Mammol 83: 1141–1150. doi:10.1644/1545-1542(2002)083<1141:COPACP>2.0.CO;2

Labbe TR, Fausch KD (2000) Dynamics of intermittent stream habitat regulate persistence of a threatened fish at multiple scales. Ecol Appl 10:1774–1791. doi:10.1890/1051-0761(2000)010[1774:DOISHR]2.0.CO;2

Lammi A, Siikamäki P, Mustajärvi (1999) Genetic diversity, population size, and fitness in central and peripheral populations of a rare plant *Lychnis viscaria*. Conserv Biol 13:1069–1078. doi:10.1046/j.1523-1739.1999.98278.x

Langella O (2002) POPULATIONS 1.2.28. Logiciel de genetique des populations. Laboratoire Populations, genetique et evolution, CNRS UPR 9034, Gif-sur-Yvette, http://www.cnrs-gif.fr/pge/

Lesica P, Allendorf FW (1995) When are peripheral populations valuable for conservation? Conserv Biol 9:753–760. doi:10.1046/j.1523-1739.1995.09040753.x

Magnuson JJ, Webster KE, Assel RA et al (1997) Potential effects of climate change on aquatic systems: Laurentian Great Lakes and Precambrian shield region. Hydrol Process 11:825–871. doi:10.1002/(SICI)1099-1085(19970630)11:8<825::AID-HYP509>3.0.CO;2-G

Mandrak NE, Crossman EJ (1992) Postglacial dispersal of freshwater fishes in Ontario. Can J Zool 70:2247–2259

Miller MP (1997) Tools for population genetic analyses (TFPGA): a Windows program for the analysis of allozyme and molecular population genetic data v.1.3, February 2000. Available via http://bioweb.usu.edu/mpmbio. Accessed 30 Nov 2007

Page RDM (2001) TREEVIEW (Win32) v. 5. URL: http://taxonomy.zoology.gla.ac.uk/rod/rod.html. Accessed 30 Nov 2007

Peakall R, Smouse PE (2006) GENALEX 6: genetic analysis in Excel. Population genetic software for teaching and research. Mol Ecol Notes 6:288–295. doi:10.1111/j.1471-8286.2005.01155.x

Piry S, Alapetite A, Cornuet JM et al (2004) GENECLASS 2: a software for genetic assignment and first generation migrants detection. J Hered 95:536–539. doi:10.1093/jhered/esh074

Quinn JF, Karr JR (1993) Habitat fragmentation and global change. In: Kareiva PM, Kingsolver JG, Huey RB (eds) Biotic interactions and global change. Sinauer and Associates, Massachusetts, pp 451–463

Rannala B, Mountain JL (1997) Detecting immigration by using multilocus genotypes. Proc Natl Acad Sci USA 94:9197–9201. doi:10.1073/pnas.94.17.9197

Raymond M, Rousset F (1995) An exact test for populations differentiation. Evol Int J Org Evol 49:1280–1283. doi:10.2307/2410454

Ricciardi A (2003) Predicting the impacts of an introduced species from its invasion history: an empirical approach applied to zebra mussel invasions. Freshw Biol 48:972–981. doi:10.1046/j.1365-2427.2003.01071.x

Schlosser IJ, Toth LA (1984) Niche relationships and population ecology of rainbow (*Etheostoma caeruleum*) and fantail (*E. flabellare*) darters in a temporally variable environment. Oikos 42:229–238. doi:10.2307/3544798

Schneider S, Roessli D, Excoffier L (2000) ARLEQUIN v. 3.0. Genetics and Biometry Laboratory, University of Geneva, Switzerland

Scott WB, Crossman EJ (1973) Freshwater fishes of Canada. Bull Fish Res Board Can 184: 966 pp

Tonnis BD (2006) Microsatellite DNA markers for the rainbow darter, *Etheostoma caeruleum* (Percidae), and their potential utility for other darter species. Mol Ecol Notes 6:230–232. doi:10.1111/j.1471-8286.2005.01203.x

Turner TF (2001) Comparative study of larval transport and gene flow in darters. Copeia 2001:766–774. doi:10.1643/0045-8511(2001)001[0766:CSOLTA]2.0.CO;2

Turner TE, Trexler JC (1998) Ecological and historical associations of gene flow in darters (Teleostei: percidae). Evol Int J Org Evol 52:1781–1801. doi:10.2307/2411349

Van Oosterhout C, Hutchinson WF, Wills DPM, Shipley P (2004) Micro-checker: software for identifying and correcting genotyping errors in microsatellite data. Mol Ecol Notes 4:535–538. doi:10.1111/j.1471-8286.2004.00684.x

Weir BS, Cockerham CC (1984) Estimating F-statistics for the analysis of population structure. Evol Int J Org Evol 38:1358–1370. doi:10.2307/2408641

Winn HE (1958) Comparative reproductive behavior and ecology of fourteen species of darters (Pisces-Percidae). Ecol Monogr 28:155–191. doi:10.2307/1942207